The Exponent Rules

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

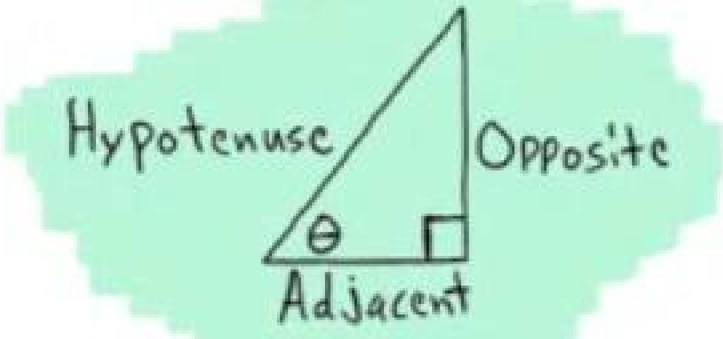
VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$



$\sin\theta = \frac{Opp}{Hyp}$ $\quad \csc\theta = \frac{Hyp}{Opp}$

$\cos\theta = \frac{Adj}{Hyp}$ $\quad \sec\theta = \frac{Hyp}{Adj}$

$\tan\theta = \frac{Opp}{Adj}$ $\quad \cot\theta = \frac{Adj}{Opp}$

Opposite Angle Identities

$\sin(-\theta) = -\sin\theta$ $\quad \cos(-\theta) = \cos\theta$ $\quad \tan(-\theta) = -\tan\theta$

$\csc(-\theta) = -\csc\theta$ $\quad \sec(-\theta) = \sec\theta$ $\quad \cot(-\theta) = -\cot\theta$

Complete Rotation and Half Rotation Identities

$\sin(\theta \pm 360°) = \sin\theta$ $\quad \sin(\theta \pm 180°) = -\sin\theta$

$\cos(\theta \pm 360°) = \cos\theta$ $\quad \cos(\theta \pm 180°) = -\cos\theta$

$\tan(\theta \pm 360°) = \tan\theta$ $\quad \csc(\theta \pm 180°) = -\csc\theta$

$\csc(\theta \pm 360°) = \csc\theta$ $\quad \sec(\theta \pm 180°) = -\sec\theta$

$\sec(\theta \pm 360°) = \sec\theta$ $\quad \tan(\theta \pm 180°) = \tan\theta$

$\cot(\theta \pm 360°) = \cot\theta$ $\quad \cot(\theta \pm 180°) = \cot\theta$

Complete Rotation Angle Pair Identities and Supplementary Angle Identities

$\sin(360° - \theta) = -\sin\theta$ $\quad \sin(180° - \theta) = \sin\theta$

$\cos(360° - \theta) = \cos\theta$ $\quad \csc(180° - \theta) = \csc\theta$

$\tan(360° - \theta) = -\tan\theta$ $\quad \cos(180° - \theta) = -\cos\theta$

$\csc(360° - \theta) = -\csc\theta$ $\quad \sec(180° - \theta) = -\sec\theta$

$\sec(360° - \theta) = \sec\theta$ $\quad \tan(180° - \theta) = -\tan\theta$

$\cot(360° - \theta) = -\cot\theta$ $\quad \cot(180° - \theta) = -\cot\theta$

Sum to Product Formulas

$\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$

$\sin A - \sin B = 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$

$\cos A + \cos B = 2\cos\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$

$\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$

Product to Sum Formulas

$\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

$\cos A\sin B = \frac{1}{2}[\sin(A+B) - \sin(A-B)]$

$\cos A\cos B = \frac{1}{2}[\cos(A+B) + \cos(A-B)]$

$\sin A\sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$

The Area of an SAS Triangle

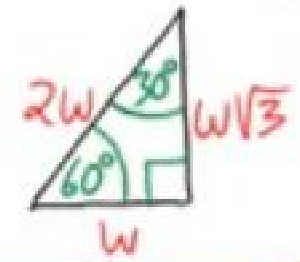
The area of a triangle is equal to the length of one side times the length of another side times the sine of the measure of the angle between the two sides divided by two.

$\text{Area} = \frac{1}{2}ab\sin C$ $\quad \text{Area} = \frac{1}{2}bc\sin A$

$\text{Area} = \frac{1}{2}ac\sin B$

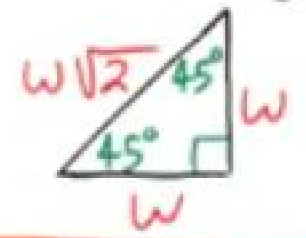
30°-60°-90° Triangle Theorem

30°-60°-90° Triangle Theorem: If the inner angles of a triangle are 30°, 60°, and 90°, and the length of the side opposite the 30° angle is w, then the length of the hypotenuse is 2w and the length of the side opposite the 60° angle is $w\sqrt{3}$.



45°-45°-90° Triangle Theorem

45°-45°-90° Triangle Theorem: If the inner angles of a triangle are 45°, 45°, and 90° and the length of the two sides opposite the 45° angles are w, then the length of the hypotenuse is $w\sqrt{2}$.



Quarter Rotation Identities

$\sin(\theta \pm 90°) = \pm\cos\theta$ $\quad \cos(\theta \pm 90°) = \mp\sin\theta$

$\csc(\theta \pm 90°) = \pm\sec\theta$ $\quad \sec(\theta \pm 90°) = \mp\csc\theta$

$\tan(\theta \pm 90°) = -\cot\theta$

$\cot(\theta \pm 90°) = -\tan\theta$

Complementary Angle Identities

$\sin(90° - \theta) = \cos\theta$ $\quad \csc(90° - \theta) = \sec\theta$

$\cos(90° - \theta) = \sin\theta$ $\quad \sec(90° - \theta) = \csc\theta$

$\tan(90° - \theta) = \cot\theta$

$\cot(90° - \theta) = \tan\theta$

Scenarios when Solving Triangles

The Law of Sines: AAS, ASA, SSA

$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

SSA → Ambiguity
i) No triangle
ii) One triangle
iii) Two triangles

The Law of Cosines: SSS, SAS

$c^2 = a^2 + b^2 - 2ab\cos C$

$b^2 = c^2 + a^2 - 2ac\cos B$

$a^2 = b^2 + c^2 - 2bc\cos A$

Heron's Formula

If a, b, and c are the side lengths of a triangle, then the area of the triangle is equal to the expression below.

$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$

where $s = \frac{a+b+c}{2}$

Reciprocal Identities

$\sin\theta = \frac{1}{\csc\theta}$ $\quad \csc\theta = \frac{1}{\sin\theta}$

$\cos\theta = \frac{1}{\sec\theta}$ $\quad \sec\theta = \frac{1}{\cos\theta}$

$\tan\theta = \frac{1}{\cot\theta}$ $\quad \cot\theta = \frac{1}{\tan\theta}$

Ratio Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

Pythagorean Identities

$\sin^2\theta + \cos^2\theta = 1$

$\tan^2\theta + 1 = \sec^2\theta$ $\quad 1 + \cot^2\theta = \csc^2\theta$

Sum and Difference Formulas

$\sin(A \pm B) = \sin A\cos B \pm \sin B\cos A$

$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B$

Double Angle Formulas

$\sin(2A) = 2\sin A\cos A$

$\cos(2A) = \cos^2 A - \sin^2 A$

$\cos(2A) = 1 - 2\sin^2 A$

$\cos(2A) = 2\cos^2 A - 1$

Half Angle Formulas

$\sin\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1-\cos A}{2}}$

$\cos\left(\frac{A}{2}\right) = \pm\sqrt{\frac{1+\cos A}{2}}$

General Forms of Trigonometric Functions

Sinusoidal functions: $y = A\sin(wx - \phi) + B$, $\quad y = A\cos(wx - \phi) + B$

$y = A\csc(wx - \phi) + B$ $\quad y = A\tan(wx - \phi) + B$

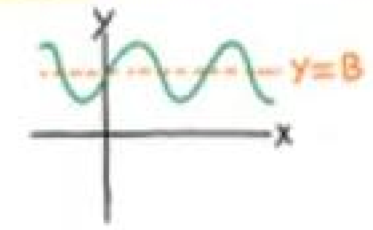
$y = A\sec(wx - \phi) + B$ $\quad y = A\cot(wx - \phi) + B$

Sinusoidal Function: A function whose graph has the same general form as the sine or cosine functions.

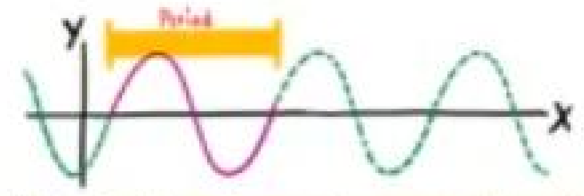
Midline: A horizontal line drawn through the middle of the graph of a trigonometric function to assist when graphing.

For the functions above, the equation of the midline is $y = B$.

Example:



Period: The number of degrees required for a trigonometric function to complete one whole cycle.



The period of the sine, cosine, cosecant, and secant functions is given by the following equation.

$P = \frac{2\pi}{w}$

The period of the tangent and cotangent functions is given by the following equation.

$P = \frac{\pi}{w}$

Phase Shift: The horizontal distance that a parent graph is translated (to the right or left).

The phase shift is given by the equation below.

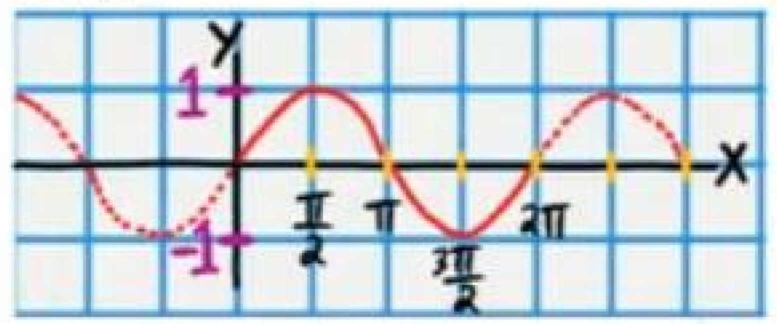
$PS = \frac{\phi}{w}$

Amplitude: The vertical distance from the midline to the highest or lowest point on a sinusoidal graph.

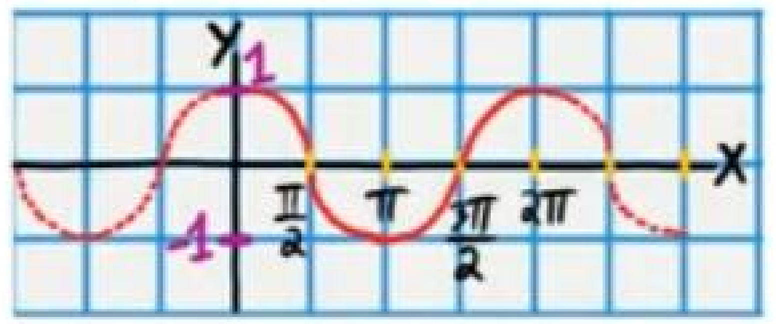
The amplitude is the magnitude (i.e. the absolute value) of the coefficient of the trigonometric function.

Example: If $y = A\sin(wx - \phi) + B$, then Amplitude $= |A|$.

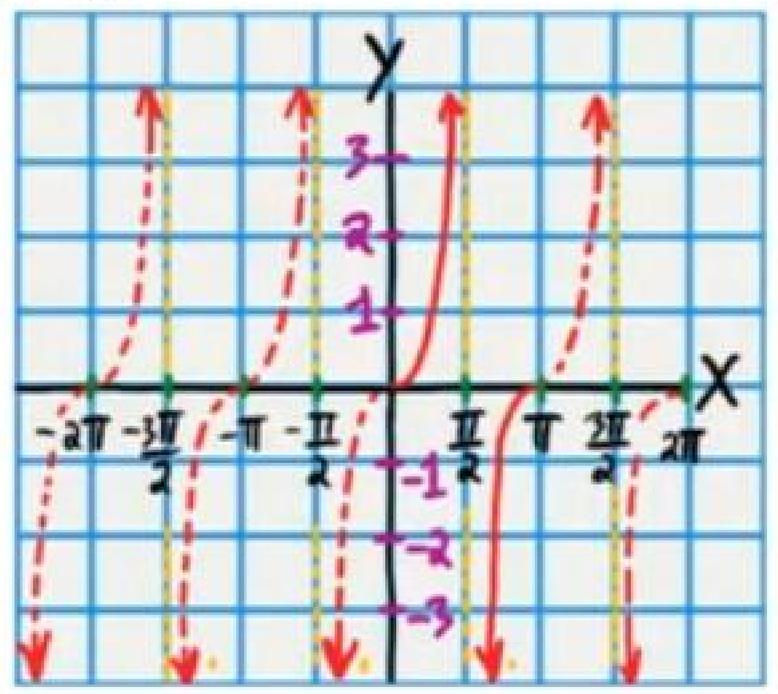
$y = \sin x$



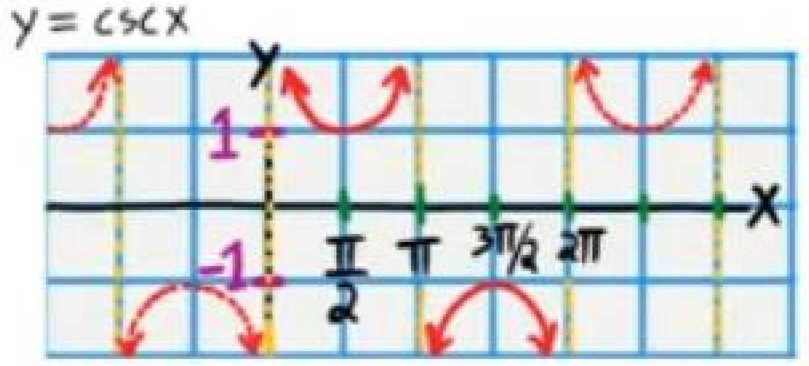
$y = \cos x$



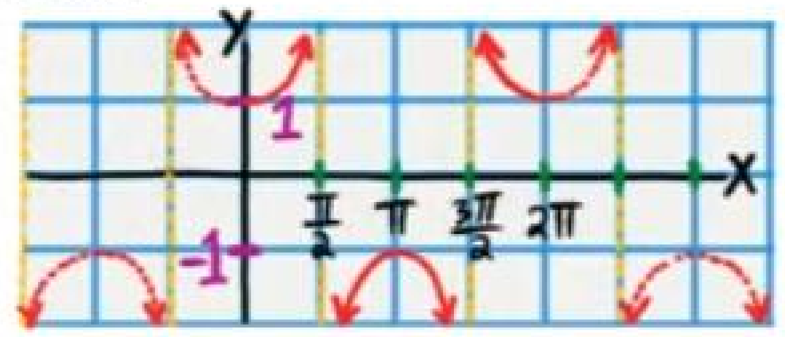
$y = \tan x$



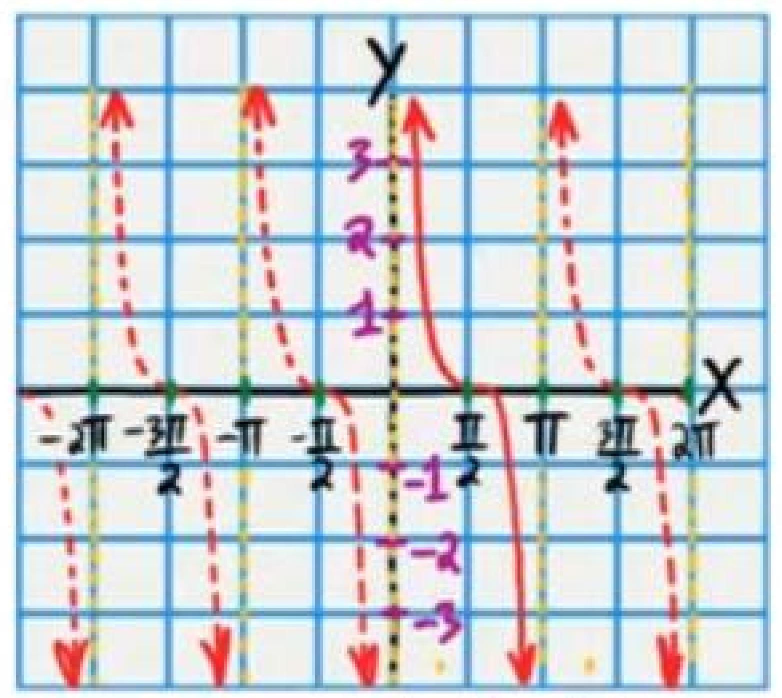
$y = \csc x$



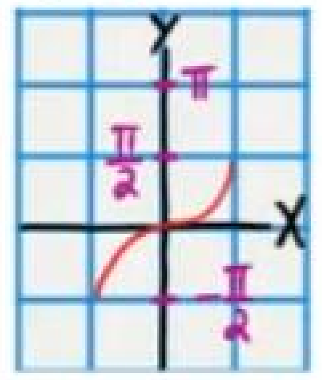
$y = \sec x$



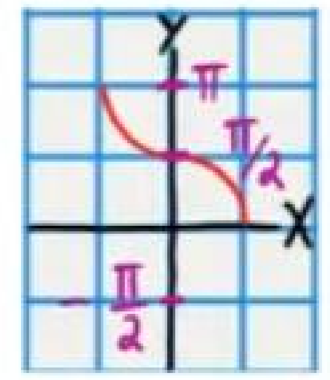
$y = \cot x$



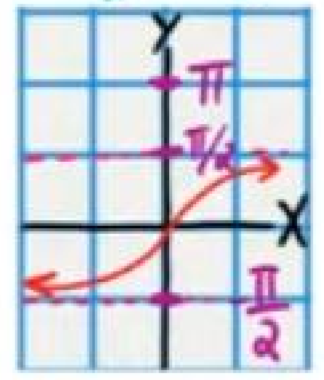
$y = \sin^{-1}x$ — Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$



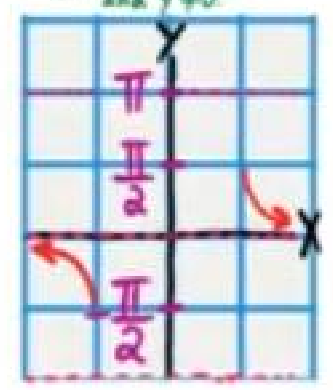
$y = \cos^{-1}x$ — $0 \le y \le \pi$



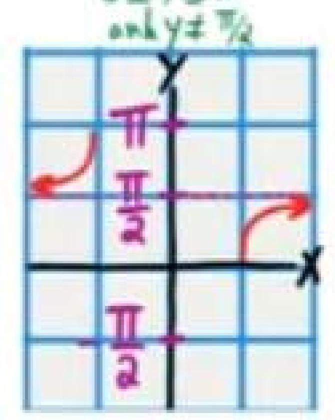
$y = \tan^{-1}x$ — $-\frac{\pi}{2} < y < \frac{\pi}{2}$



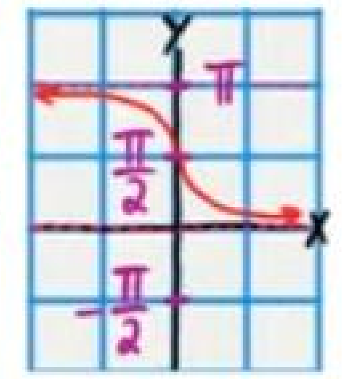
$y = \csc^{-1}x$ — Range: $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ and $y \ne 0$.



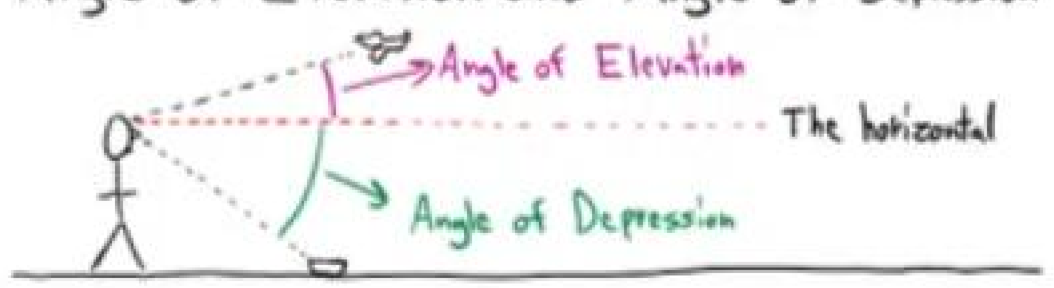
$y = \sec^{-1}x$ — $0 \le y \le \pi$ and $y \ne \frac{\pi}{2}$



$y = \cot^{-1}x$ — $0 < y < \pi$



Angle of Elevation and Angle of Depression



Converting from Degrees to Radians

degree measure $\times \frac{\pi \text{ radians}}{180 \text{ degrees}}$ = radian measure

Arc Length

$s = \theta r$ *This formula can be applied with radian units only

The Area of a Sector

$A = \frac{\theta}{2}r^2$ *This formula can be applied with radian units only.

Converting from Radians to Degrees

Radian measure $\times \frac{180 \text{ degrees}}{\pi \text{ radians}}$ = degree measure

If an object moves s units on a circular path in t units of time, then V is defined as the linear velocity according to the equation below.

$V = \frac{s}{t}$

If an object on a circle rotates θ angle units in t units of time, then ω is defined as the angular velocity according to the equation below.

$\omega = \frac{\theta}{t}$

$V = r\omega$ *This formula can be applied with radian units only.

Revolution: A complete rotation.

Revolutions per minute = rpm

1 Revolution = 2π radians = 360°

Name
Date
Course

Finding the Equation of a Tangent Line

Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c)=0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n)=nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)]=cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x)+g(x)]=f'(x)+g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x)-g(x)]=f'(x)-g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)]=f'(x)g(x)+f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]=\frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$

The Chain Rule viii) If $h(x)=f(g(x))$, then

$$\boxed{h'(x)=f'(g(x))g'(x)}$$

Rule for Exponential Functions ix) $\frac{d}{dx}(a^x)=a^x\ln a$, "a" is a constant and $a>0$.

Rule for base "e" Exponential Functions x) $\frac{d}{dx}(e^x)=e^x$

Rule for Logarithmic Functions xi) $\frac{d}{dx}(\log_a x)=\frac{1}{x\ln a}$, "a" is a constant and $a>0$.

Rule for the Natural Logarithm Function xii) $\frac{d}{dx}(\ln x)=\frac{1}{x}$

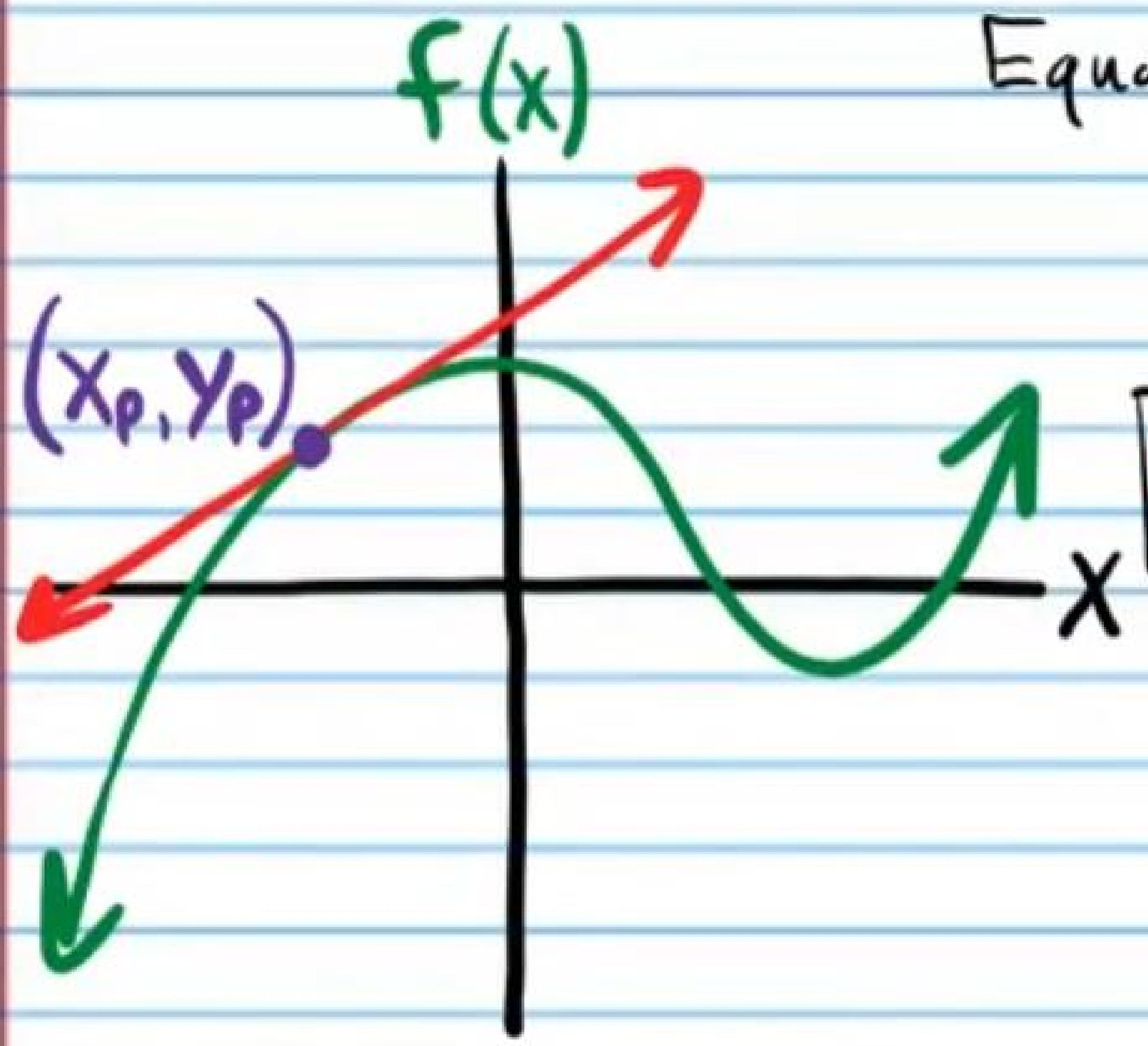
Rule for Composite Natural Logarithm Functions xiii) $\frac{d}{dx}\{\ln[g(x)]\}=\frac{g'(x)}{g(x)}$

Trigonometric Differentiation Formulas

I) $\frac{d}{dx}(\sin x)=\cos x$ II) $\frac{d}{dx}(\cos x)=-\sin x$

III) $\frac{d}{dx}(\csc x)=-\csc x\cot x$ IV) $\frac{d}{dx}(\sec x)=\sec x\tan x$

V) $\frac{d}{dx}(\tan x)=\sec^2 x$ VI) $\frac{d}{dx}(\cot x)=-\csc^2 x$



Equation of the Tangent Line

$y - y_p = m(x - x_p)$

$$\boxed{y - y_p = f'(x_p)(x - x_p)}$$

Find an equation of the line tangent to the graph of each function at the indicated point on the graph.

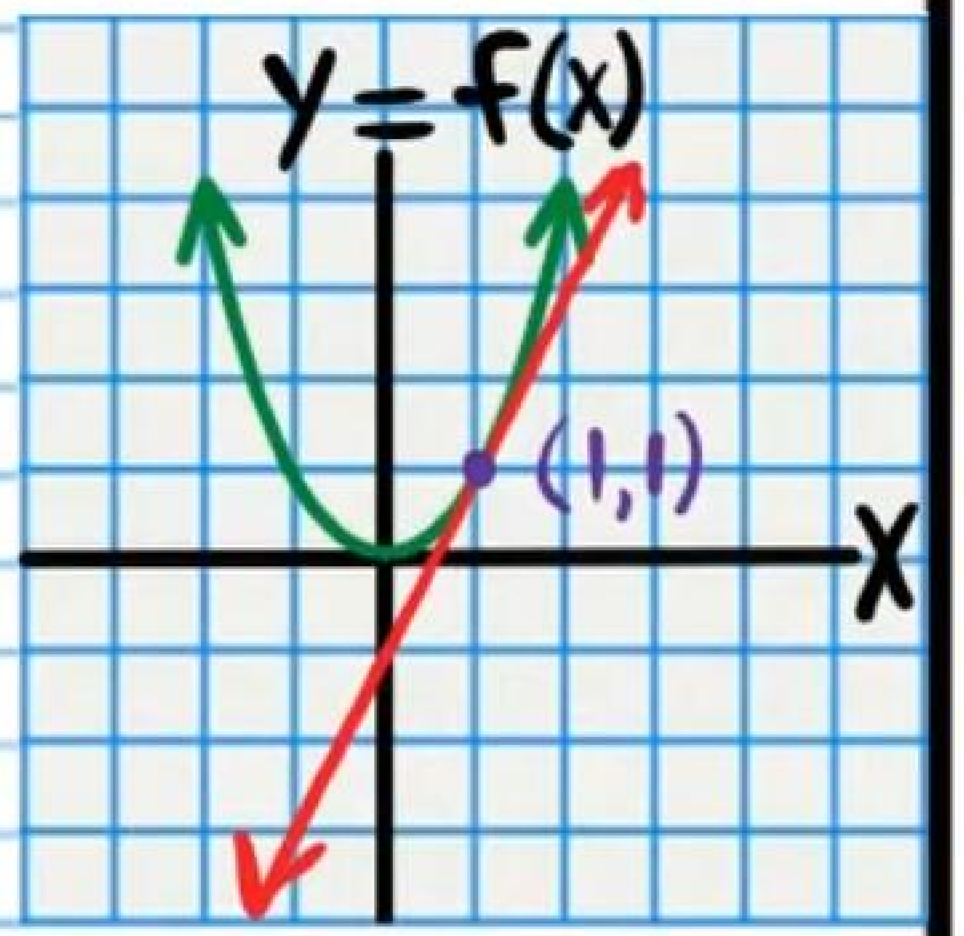
① $f(x) = x^2$, $(1,1)$

Finding the Slope of the Tangent Line

$f'(x) = 2x$

$f'(1) = 2(1)$

$f'(1) = 2$



Writing the Equation of the Tangent Line

$y - y_p = m(x - x_p)$

$$\boxed{y - 1 = 2(x - 1)}$$

② $f(x) = 2^{-x}$, $(-1,2)$

Finding the Slope of the Tangent Line

$f'(x) = 2^{-x} \ln 2 \frac{d}{dx}(-x)$

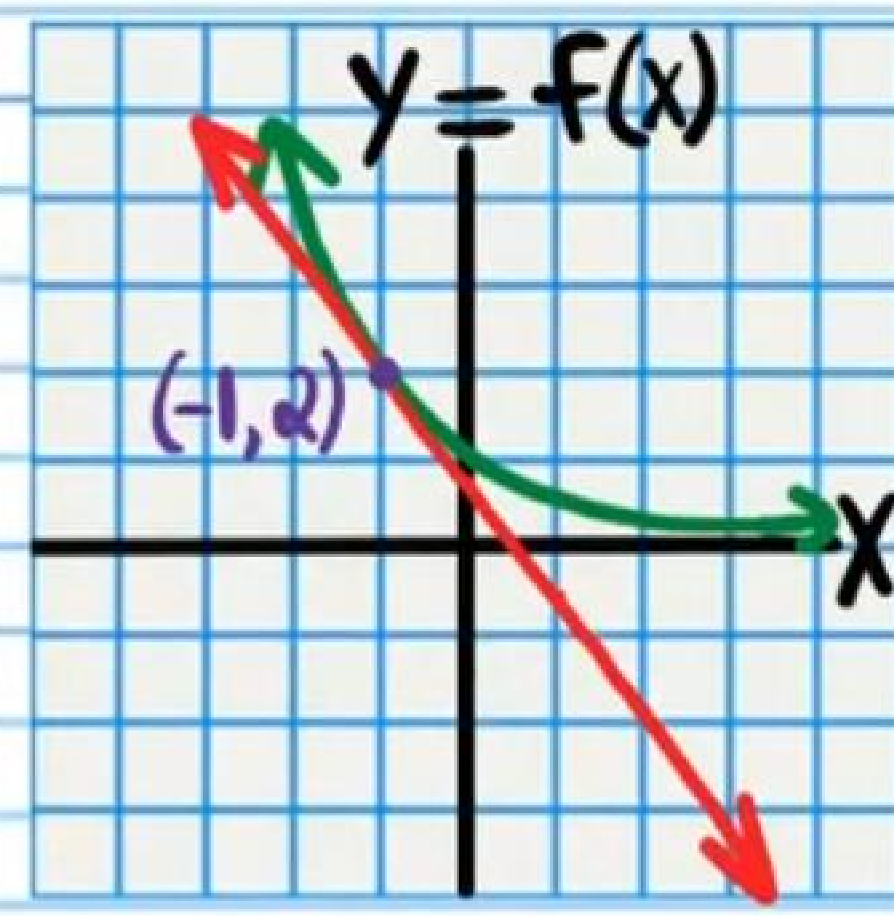
$f'(x) = 2^{-x} \ln 2 \, (-1)$

$f'(x) = -2^{-x} \ln 2$

$f'(-1) = -2^{-(-1)} \ln 2$

$f'(-1) = -2^{1} \ln 2$

$f'(-1) = -2 \ln 2$



Writing the Equation of the Tangent Line

$y - y_p = m(x - x_p)$

$y - 2 = -2\ln 2(x - (-1))$

$$\boxed{y - 2 = -2\ln 2(x + 1)}$$

③ $f(x) = \frac{-2}{x-2}$, $(-2, \frac{1}{2})$

Finding the Slope of the Tangent Line

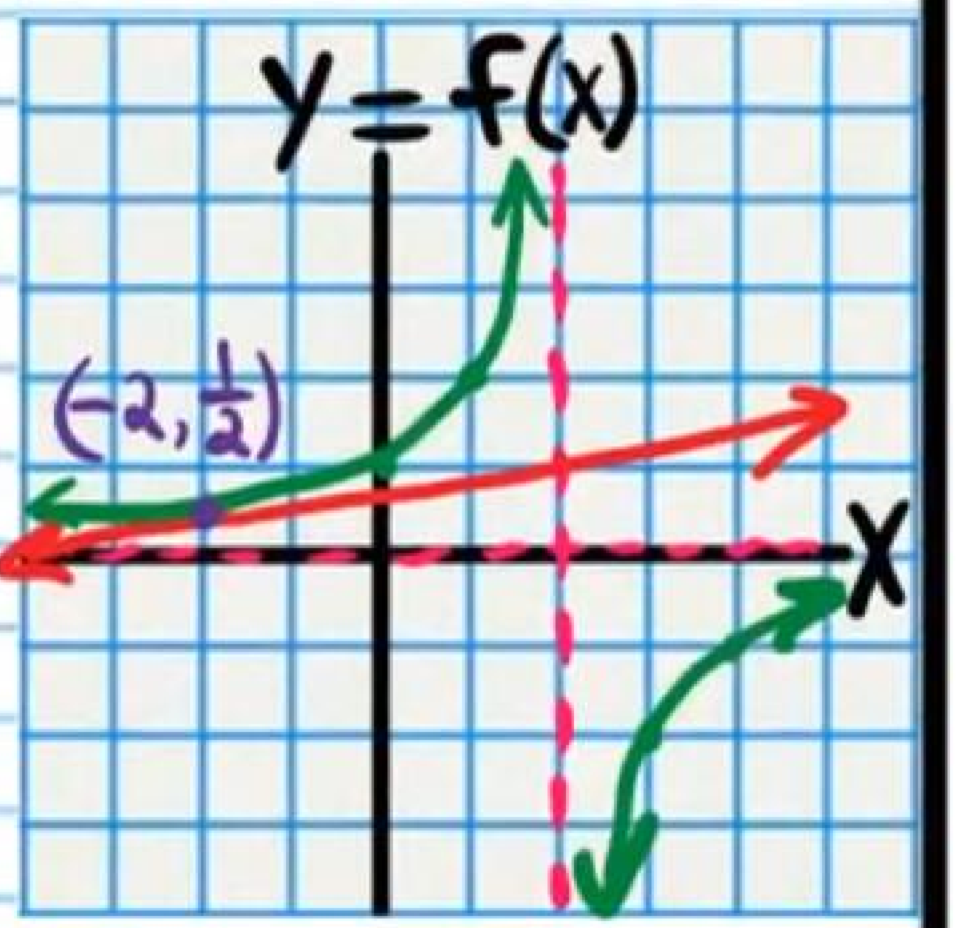
$$f'(x) = \frac{\left[\frac{d}{dx}(-2)\right](x-2) - (-2)\frac{d}{dx}(x-2)}{(x-2)^2}$$

$$f'(x) = \frac{0(x-2) + 2(1)}{(x-2)^2}$$

$$f'(x) = \frac{2}{(x-2)^2}$$

$$f'(-2) = \frac{2}{(-2-2)^2}$$

$$= \frac{1}{8}$$



Writing the Equation of the Tangent Line

$$y - y_p = m(x - x_p)$$

$$y - \frac{1}{2} = \frac{1}{8}(x - (-2))$$

$$\boxed{y - \frac{1}{2} = \frac{1}{8}(x+2)}$$

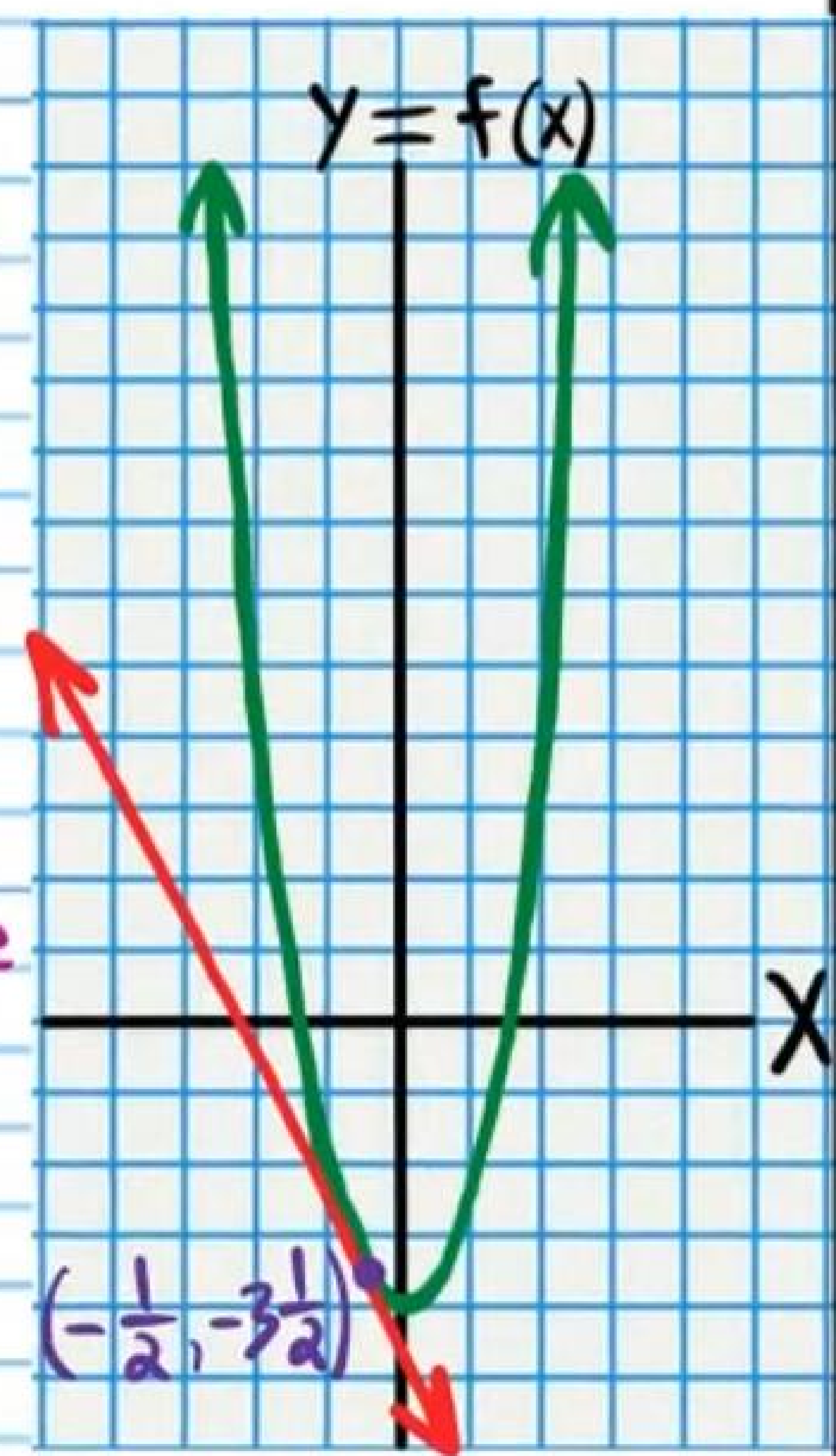
④ $f(x) = 2x^2 - 4$, $(-\frac{1}{2}, -3\frac{1}{2})$

Finding the Slope of the Tangent Line

$$f'(x) = 4x$$

$$f'(-\tfrac{1}{2}) = 4(-\tfrac{1}{2})$$

$$f'(-\tfrac{1}{2}) = -2$$



Writing the Equation of the Tangent Line

$$y - y_p = m(x - x_p)$$

$$y - (-3\tfrac{1}{2}) = -2(x - (-\tfrac{1}{2}))$$

$$\boxed{y + 3\tfrac{1}{2} = -2(x + \tfrac{1}{2})}$$

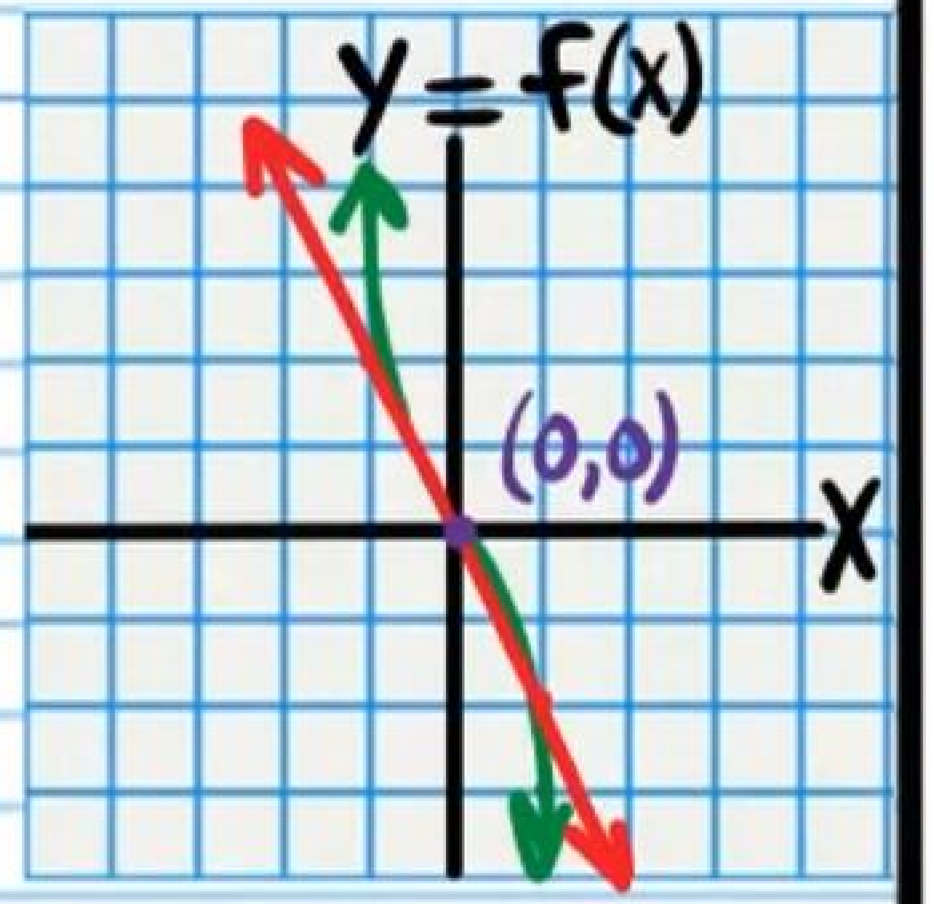
⑤ $f(x) = -x^3 - 2x$, $(0,0)$

Finding the Slope of the Tangent Line

$$f'(x) = -3x^2 - 2$$

$$f'(0) = -3(0)^2 - 2$$

$$f'(0) = -2$$



Writing the Equation of the Tangent Line

$$y - y_p = m(x - x_p)$$

$$y - 0 = -2(x - 0)$$

$$\boxed{y = -2x}$$

Rule for Composite Natural Logarithm Functions

xiii) $\frac{d}{dx}\{\ln[g(x)]\} = \frac{g'(x)}{g(x)}$

⑥ $f(x) = \ln(x-1)$ at $(4, \ln 3)$

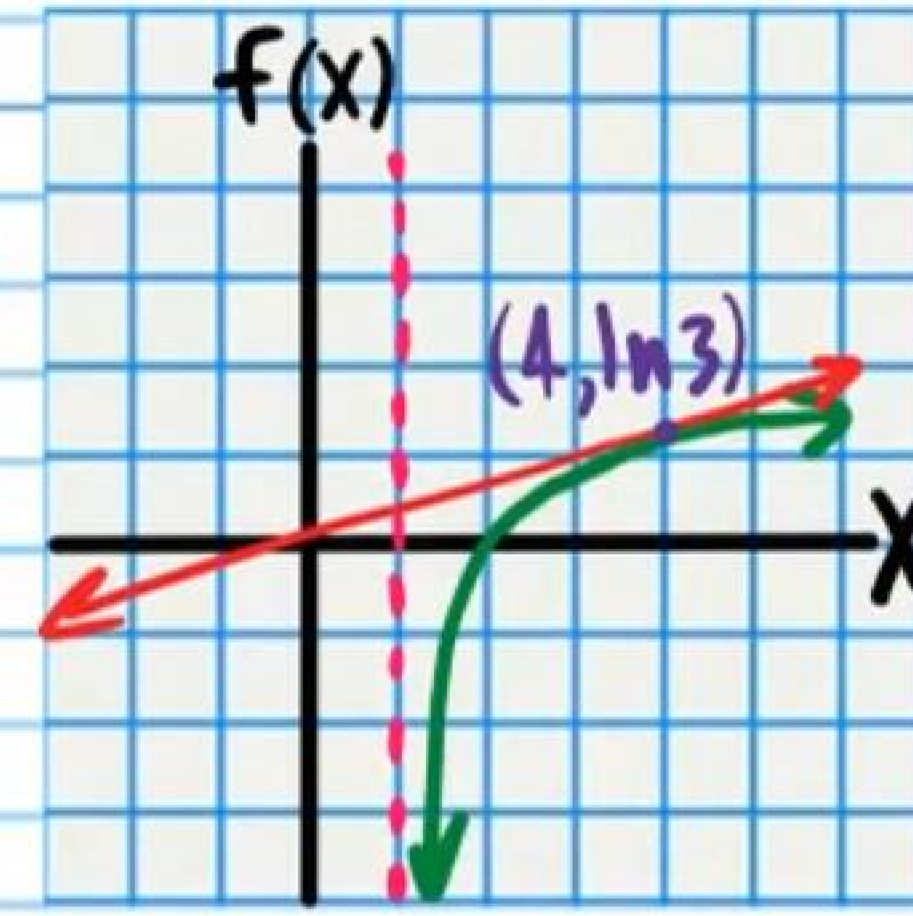
Finding the Slope of the Tangent Line

$$f'(x) = \frac{\frac{d}{dx}(x-1)}{x-1}$$

$$f'(x) = \frac{1}{x-1}$$

$$f'(4) = \frac{1}{4-1}$$

$$f'(4) = \frac{1}{3}$$



Writing the Equation of the Tangent Line

$$y - y_p = m(x - x_p)$$

$$\boxed{y - \ln 3 = \frac{1}{3}(x - 4)}$$

Find an equation of the line tangent to the graph of each function at the indicated x-value.

⑦ $f(x) = \frac{6}{x^2} - x^3$ at $x = -1$.

Finding the y-value of the point of tangency.

$f(-1) = \frac{6}{(-1)^2} - (-1)^3$

$f(-1) = \frac{6}{1} - (-1)$

$= 7$

$(-1, 7)$

Finding the slope of the tangent line.

$f(x) = 6x^{-2} - x^3$

$f(x) = -12x^{-3} - 3x^2$

$f(x) = -\frac{12}{x^3} - 3x^2$

$f(-1) = -\frac{12}{(-1)^3} - 3(-1)^2$

$f(-1) = -\frac{12}{-1} - 3(1)$

$= 12 - 3$

$= 9$

Finding the Equation

$y - y_1 = m(x - x_1)$

$y - 7 = 9(x - (-1))$

$\boxed{y - 7 = 9(x+1)}$

⑧ $f(x) = \sqrt[3]{5x}$ at $x = 25$

Finding the y-value of the point of tangency.

$f(25) = \sqrt[3]{5(25)}$

$= \sqrt[3]{125}$

$= 5$

$(25, 5)$

Finding the Equation

$y - y_p = m(x - x_p)$

$\boxed{y - 5 = \frac{1}{15}(x - 25)}$

Finding the slope of the tangent line.

Method 1

$f(x) = (5x)^{\frac{1}{3}}$

$f'(x) = \frac{1}{3}(5x)^{-\frac{2}{3}} \frac{d}{dx}(5x)$

$f'(x) = \frac{1}{3(5x)^{2/3}} \, 5$

$f'(x) = \frac{5}{3\sqrt[3]{(5x)^2}}$

$f'(25) = \frac{5}{3\sqrt[3]{(5 \cdot 25)^2}}$

$= \frac{5}{3\sqrt[3]{125^2}}$

$= \frac{5}{3(\sqrt[3]{125})^2}$

$= \frac{5}{3(5)^2} = \frac{1}{15}$

Method 2

$f(x) = \sqrt[3]{5}\,\sqrt[3]{x}$

$f(x) = \sqrt[3]{5}\, x^{\frac{1}{3}}$

⑨ $f(x) = 4e^{2x} + 10$ at $x = 0.$

Finding the y-value of the point of tangency.

$f(0) = 4e^{2(0)} + 10$

$f(0) = 4e^{0} + 10$

$= 4(1) + 10$

$= 14$

$(0, 14)$

Finding the slope of the tangent line.

$f'(x) = 4e^{2x} \frac{d}{dx}(2x) + 0$

$f'(x) = 4e^{2x}(2)$

$f'(x) = 8e^{2x}$

$f'(0) = 8e^{2(0)}$

$= 8e^{0}$

$= 8(1)$

$= 8$

Finding the Equation

$y - y_p = m(x - x_p)$

$y - 14 = 8(x - 0)$

$\boxed{y - 14 = 8x}$

⑩ $f(x) = \frac{3}{\sqrt{x}} + 5$ at $x = 4.$

Finding the y-value of the point of tangency.

$f(x) = \frac{3}{\sqrt{4}} + 5$

$= \frac{3}{2} + 5$

$= 6\frac{1}{2}$

$(4, 6\frac{1}{2})$

Finding the slope of the tangent line.

$f(x) = 3x^{-\frac{1}{2}} + 5$

$f'(x) = -\frac{3}{2}x^{-\frac{3}{2}} + 0$

$f'(x) = -\frac{3}{2\sqrt{x^3}}$

$f'(4) = -\frac{3}{2\sqrt{4^3}}$

$f'(4) = -\frac{3}{2\sqrt{64}}$

$= -\frac{3}{2(8)}$

$= -\frac{3}{16}$

Finding the Equation

$y - y_p = m(x - x_p)$

$\boxed{y - 6\frac{1}{2} = -\frac{3}{16}(x - 4)}$

(11) $f(x) = -2x^4 + x^3 - 2x + 4$ at $x = -2$.

Finding the y-value of the point of tangency.

$$f(-2) = -2(-2)^4 + (-2)^3 - 2(-2) + 4$$

$$= -2(16) - 8 + 4 + 4$$

$$= -32$$

$$(-2, -32)$$

Finding the slope of the tangent line.

$$f'(x) = -8x^3 + 3x^2 - 2$$

$$f'(-2) = -8(-2)^3 + 3(-2)^2 - 2$$

$$= -8(-8) + 3(4) - 2$$

$$= 64 + 12 - 2$$

$$= 76 - 2$$

$$= 74$$

Finding the Equation

$$y - y_p = m(x - x_p)$$

$$y - (-32) = 74(x - (-2))$$

$$\boxed{y + 32 = 74(x + 2)}$$

$$f(x) = \frac{1}{3}e^{x^2} - 7$$

(12) $f(x) = \dfrac{e^{x^2}}{3} - 7$ at $x = -3$.

Finding the y-value of the point of tangency.

$$f(-3) = \frac{e^{(-3)^2}}{3} - 7$$

$$= \frac{e^9}{3} - 7$$

$$\left(-3, \frac{e^9}{3} - 7\right)$$

Finding the slope of the tangent line.

$$f(x) = \frac{1}{3}e^{x^2} - 7$$

$$f'(x) = \frac{1}{3}e^{x^2}\frac{d}{dx}(x^2) - 0$$

$$f'(x) = \frac{1}{3}e^{x^2}(2x)$$

$$f'(x) = \frac{2}{3}xe^{x^2}$$

$$f'(-3) = \frac{2}{3}(-3)e^{(-3)^2}$$

$$f'(-3) = -2e^9$$

Finding the Equation

$$y - y_p = m(x - x_p)$$

$$y - \left(\frac{e^9}{3} - 7\right) = -2e^9(x - (-3))$$

$$\boxed{y - \frac{e^9}{3} + 7 = -2e^9(x + 3)}$$

Use implicit differentiation to find an equation of the line tangent to the graph of each relation at the indicated point.

(13) $x^2+y^2=9, \quad (2,\sqrt{5})$

Finding the Slope of the Tangent Line

$$2x+2y^{2-1}y'=0$$

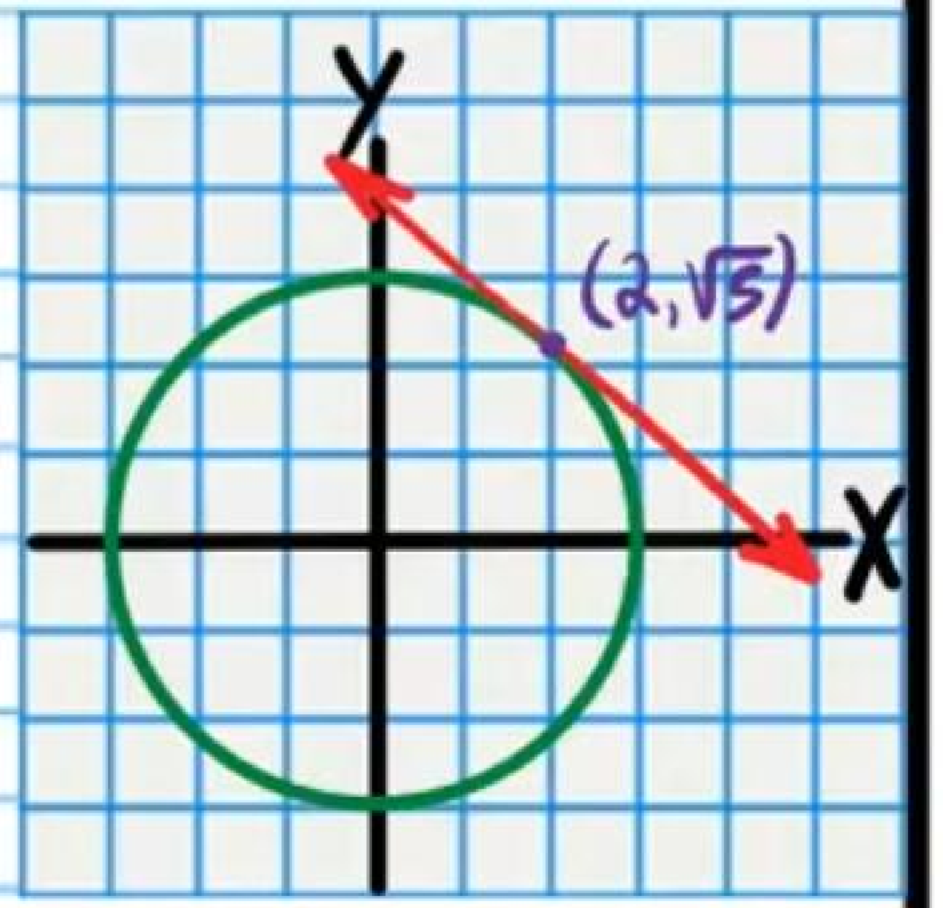
$$2x+2yy'=0$$

$$2yy'=-2x$$

$$y'=\frac{-2x}{2y}$$

$$y'=-\frac{x}{y}$$

$$y'=-\frac{2}{\sqrt{5}}$$



Writing the Equation of the Tangent Line

$$y-y_p=m(x-x_p)$$

$$\boxed{y-\sqrt{5}=-\frac{2}{\sqrt{5}}(x-2)}$$

(14) $y^3=x^2+18, \quad (3,3)$

Finding the Slope of the Tangent Line

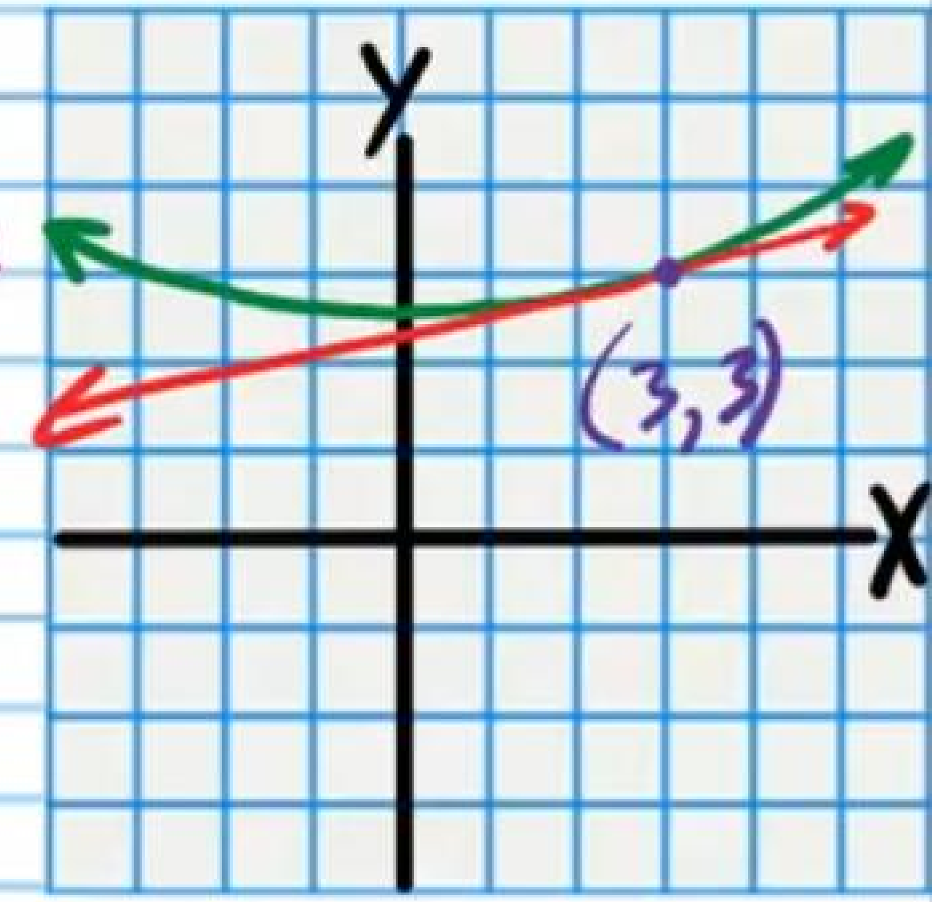
$$3y^2y'=2x+0$$

$$y'=\frac{2x}{3y^2}$$

$$y'=\frac{2(3)}{3(3)^2}$$

$$y'=\frac{6}{27}$$

$$y'=\frac{2}{9}$$



Writing the Equation of the Tangent Line

$$y-y_p=m(x-x_p)$$

$$\boxed{y-3=\frac{2}{9}(x-3)}$$

(15) $y^3+4x^2=y+1, \quad \left(-\frac{1}{2},0\right)$

Finding the Slope of the Tangent Line

$$3y^2y'+8x=y'+0$$

$$3y^2y'-y'=-8x$$

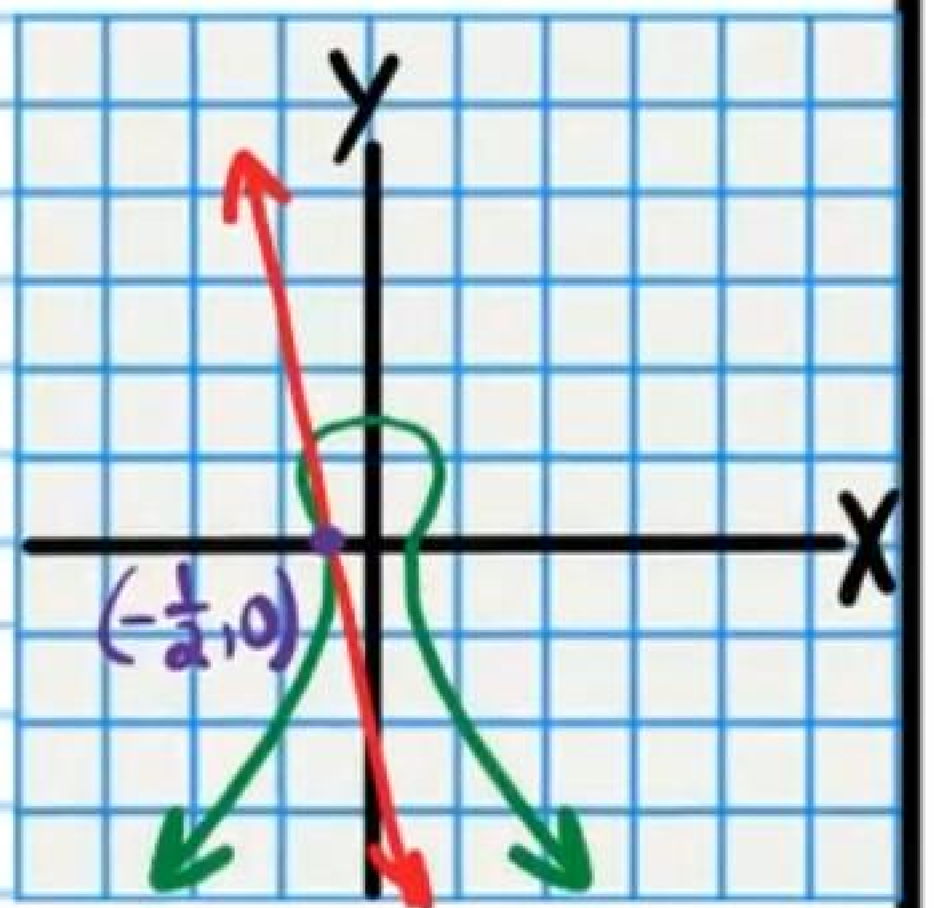
$$y'(3y^2-1)=-8x$$

$$y'=\frac{-8x}{3y^2-1}$$

$$y'=\frac{-8\left(-\frac{1}{2}\right)}{3(0)^2-1}$$

$$y'=\frac{4}{-1}$$

$$y'=-4$$



Writing the Equation of the Tangent Line

$$y-y_p=m(x-x_p)$$

$$y-0=-4\left(x-\left(-\frac{1}{2}\right)\right)$$

$$\boxed{y=-4\left(x+\frac{1}{2}\right)}$$

$$\frac{1}{9}x^2+\frac{1}{4}y^2=1$$

(16) $\frac{x^2}{9}+\frac{y^2}{4}=1, \quad \left(-1,\frac{4\sqrt{2}}{3}\right)$

Finding the Slope of the Tangent Line

$$\frac{2}{9}x+\frac{1}{2}yy'=0$$

$$\frac{1}{2}yy'=-\frac{2}{9}x$$

$$yy'=-\frac{4}{9}x$$

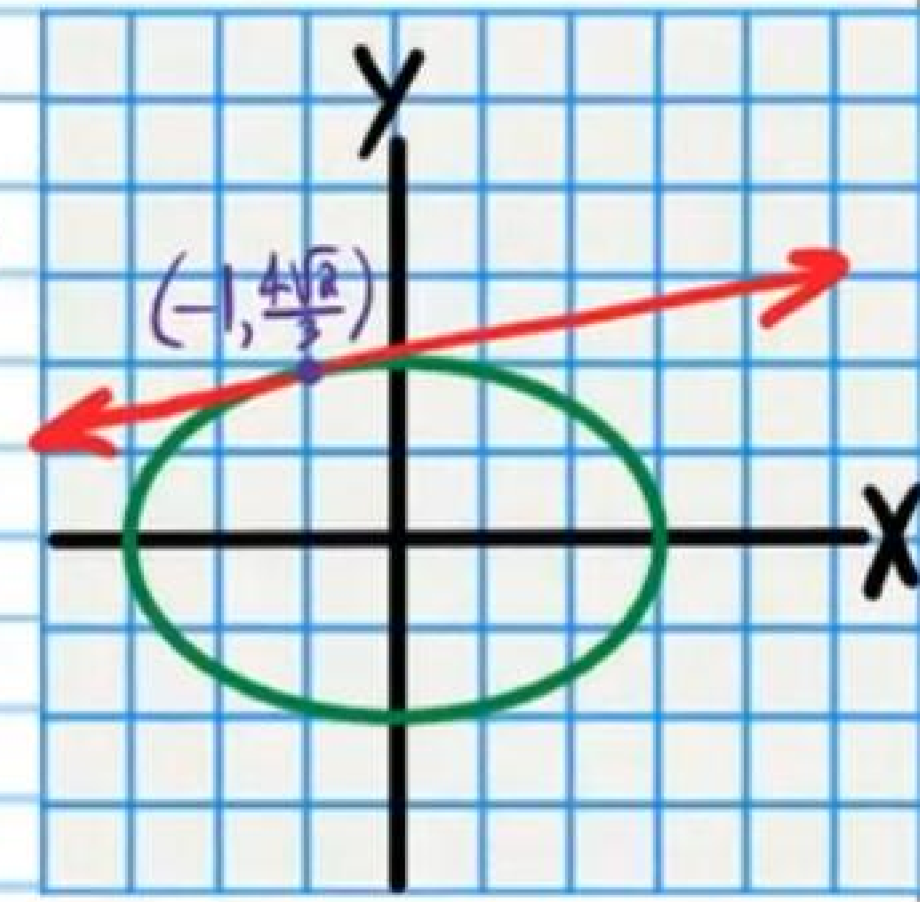
$$y'=-\frac{4x}{9y}$$

$$y'=\frac{-4(-1)}{9\left(\frac{4\sqrt{2}}{3}\right)}$$

$$=\frac{4}{12\sqrt{2}}$$

$$=\frac{1}{3\sqrt{2}}$$

$$=\frac{\sqrt{2}}{6}$$



Writing the Equation of the Tangent Line

$$y-y_p=m(x-x_p)$$

$$y-\frac{4\sqrt{2}}{3}=\frac{\sqrt{2}}{6}(x-(-1))$$

$$\boxed{y-\frac{4\sqrt{2}}{3}=\frac{\sqrt{2}}{6}(x+1)}$$

Use implicit differentiation to find an equation of the line tangent to the graph of each relation at the indicated x-value or y-value.

(17) $y^3 = 9x^4 - x^2$, $x = 1$

Finding the y-value of the point of tangency.

$$y^3 = 9(1)^4 - (1)^2$$

$$y^3 = 8$$

$$y = 2$$

$$(1, 2)$$

Finding the slope of the tangent line.

$$3y^2 y' = 36x^3 - 2x$$

$$y' = \frac{36x^3 - 2x}{3y^2}$$

$$y' = \frac{36(1)^3 - 2(1)}{3(2)^2}$$

$$y' = \frac{34}{12}$$

$$y' = \frac{17}{6}$$

Finding the Equation

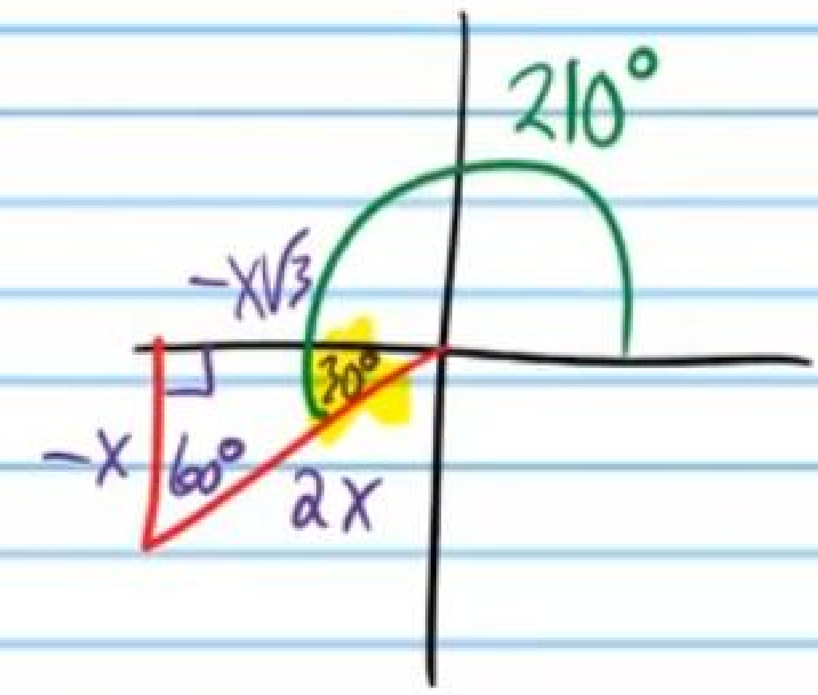
$$y - y_p = m(x - x_p)$$

$$\boxed{y - 2 = \frac{17}{6}(x - 1)}$$

(18) $x^3 = \sin y$, $y = \frac{7\pi}{6}$

Finding the x-value of the point of tangency.

$$x^3 = \sin\left(\frac{7\pi}{6}\right)$$



$$x^3 = -\frac{1}{2}$$

$$x = \sqrt[3]{-\frac{1}{2}}$$

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

$$\left(\sqrt[3]{-\tfrac{1}{2}}, \frac{7\pi}{6}\right)$$

Finding the slope of the tangent line.

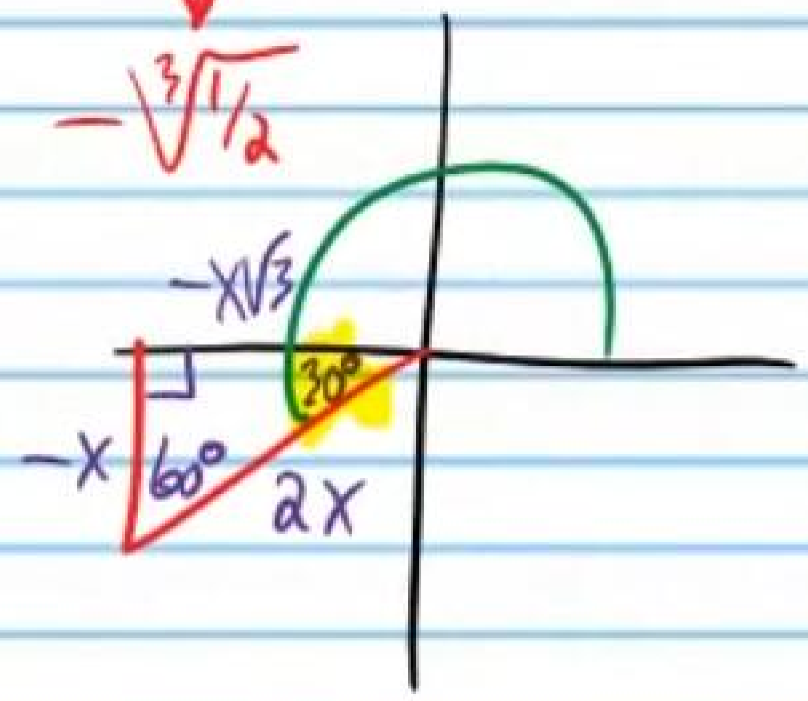
$$x^3 = \sin y$$

$$3x^2 = y' \cos y$$

$$y' = \frac{3x^2}{\cos y}$$

$$y' = \frac{3\left(\sqrt[3]{-1/2}\right)^2}{\cos \frac{7\pi}{6}} \qquad \left(\sqrt[3]{-1/2}, \frac{7\pi}{6}\right)$$

$$-\sqrt[3]{1/2}$$



$$y' = \frac{3\left(\sqrt[3]{-1/2}\right)^2}{-\frac{\sqrt{3}}{2}}$$

$$y' = \frac{3\left(\sqrt[3]{-1/2}\right)^2 \cdot 2}{-\frac{\sqrt{3}}{2} \cdot 2}$$

$$= \frac{6\left(\sqrt[3]{-1/2}\right)^2 \cdot \sqrt{3}}{-\sqrt{3} \cdot \sqrt{3}}$$

$$= \frac{6\sqrt{3}\left(\sqrt[3]{-1/2}\right)^2}{-3} = -2\sqrt{3}\sqrt[3]{\left(1/2\right)^2} = -2\sqrt{3}\sqrt[3]{1/4}$$

Finding the Equation

$$y - y_p = m(x - x_p)$$

$$y - \frac{7\pi}{6} = -2\sqrt{3}\sqrt[3]{1/4}\left(x - \left(-\sqrt[3]{1/2}\right)\right)$$

$$\boxed{y - \frac{7\pi}{6} = -2\sqrt{3}\sqrt[3]{1/4}\left(x + \sqrt[3]{1/2}\right)}$$

---

(19) $\sqrt{y} + 32 = 4x^2; \quad x = -3$

Finding the y-value of the point of tangency.

$$\sqrt{y} + 32 = 4(-3)^2$$

$$\sqrt{y} + 32 = 36$$

$$\sqrt{y} = 4$$

$$y = 16$$

$$(-3, 16)$$

Finding the slope of the tangent line.

$$y^{1/2} + 32 = 4x^2$$

$$\frac{1}{2}y^{-\frac{1}{2}}y' + 0 = 8x$$

$$\frac{y'}{2y^{1/2}} = 8x$$

$$\frac{y'}{2\sqrt{y}} = 8x$$

$$y' = 16x\sqrt{y}$$

$$y' = 16(-3)\sqrt{16}$$

$$y' = 16(-3)(4)$$

$$y' = -192$$

Finding the Equation

$$y - y_p = m(x - x_p)$$

$$y - 16 = -192(x - (-3))$$

$$\boxed{y - 16 = -192(x + 3)}$$

(20) $53x^2 = 2y^3 - x^3, \quad x = 1$

Finding the y-value of the point of tangency.

$$53(1)^2 = 2y^3 - 1^3$$
$$53 = 2y^3 - 1$$
$$54 = 2y^3$$
$$27 = y^3$$
$$y = 3$$
$$(1, 3)$$

Finding the slope of the tangent line.

$$106x = 6y^2y' - 3x^2$$
$$3x^2 + 106x = 6y^2y'$$
$$\frac{3x^2 + 106x}{6y^2} = y'$$
$$\frac{3(1)^2 + 106(1)}{6(3)^2} = y'$$
$$\frac{109}{54} = y'$$

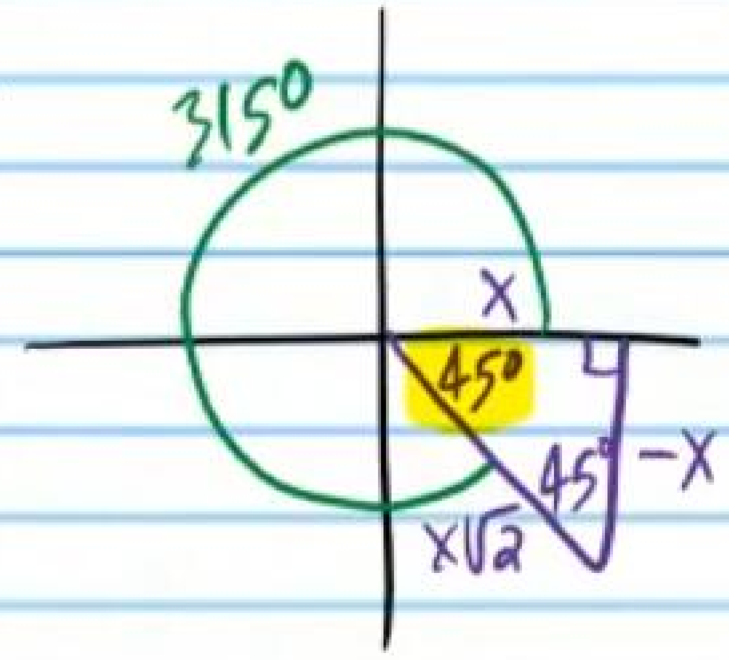
Finding the Equation

$$y - y_p = m(x - x_p)$$
$$\boxed{y - 3 = \frac{109}{54}(x - 1)}$$

(21) $\sqrt{x} = \cos y, \quad y = \frac{7\pi}{4}$

Finding the x-value of the point of tangency.

$$\sqrt{x} = \cos\left(\frac{7\pi}{4}\right)$$
$$\sqrt{x} = \frac{1}{\sqrt{2}}$$



$$(\sqrt{x})^2 = \left(\frac{1}{\sqrt{2}}\right)^2$$
$$x = \frac{1}{2}$$
$$\left(\frac{1}{2}, \frac{7\pi}{4}\right)$$

Finding the slope of the tangent line.

$$x^{\frac{1}{2}} = \cos y$$
$$\frac{1}{2}x^{-\frac{1}{2}} = -y'\sin y$$
$$\frac{1}{2\sqrt{x}} = -y'\sin y$$
$$y' = -\frac{1}{2\sqrt{x}\sin y}$$
$$y' = -\frac{1}{2\sqrt{1/2}\,\sin(7\pi/4)}$$
$$y' = -\frac{1}{2\sqrt{1/2}\,(-1/\sqrt{2})}$$
$$= \frac{1}{2\,\frac{1}{\sqrt{2}}\,\frac{1}{\sqrt{2}}}$$
$$= \frac{1}{2\left(\frac{1}{2}\right)} = 1$$

Finding the Equation

$$y - y_p = m(x - x_p)$$
$$y - \frac{7\pi}{4} = 1\left(x - \frac{1}{2}\right)$$
$$\boxed{y - \frac{7\pi}{4} = x - \frac{1}{2}}$$

(22) $2x^2-33=\frac{1}{y}, \quad x=-4$

Finding the y-value of the point of tangency.

$$2(-4)^2-33=\frac{1}{y}$$

$$2(16)-33=\frac{1}{y}$$

$$-1=\frac{1}{y}$$

$$y(-1)=\frac{1}{y}\cdot y$$

$$-y=1$$

$$y=-1$$

$$(-4,-1)$$

Finding the slope of the tangent line.

$$2x^2-33=y^{-1}$$

$$4x-0=-y^{-2}y'$$

$$4x=-y^{-2}y'$$

$$4x=-\frac{y'}{y^2}$$

$$-4xy^2=y'$$

$$-4(-4)(-1)^2=y'$$

$$16=y'$$

Finding the Equation

$$y-y_p=m(x-x_p)$$

$$y-(-1)=16(x-(-4))$$

$$\boxed{y+1=16(x+4)}$$

Rule for Composite Natural Logarithm Functions xiii) $\frac{d}{dx}\{\ln[g(x)]\}=\frac{g'(x)}{g(x)}$

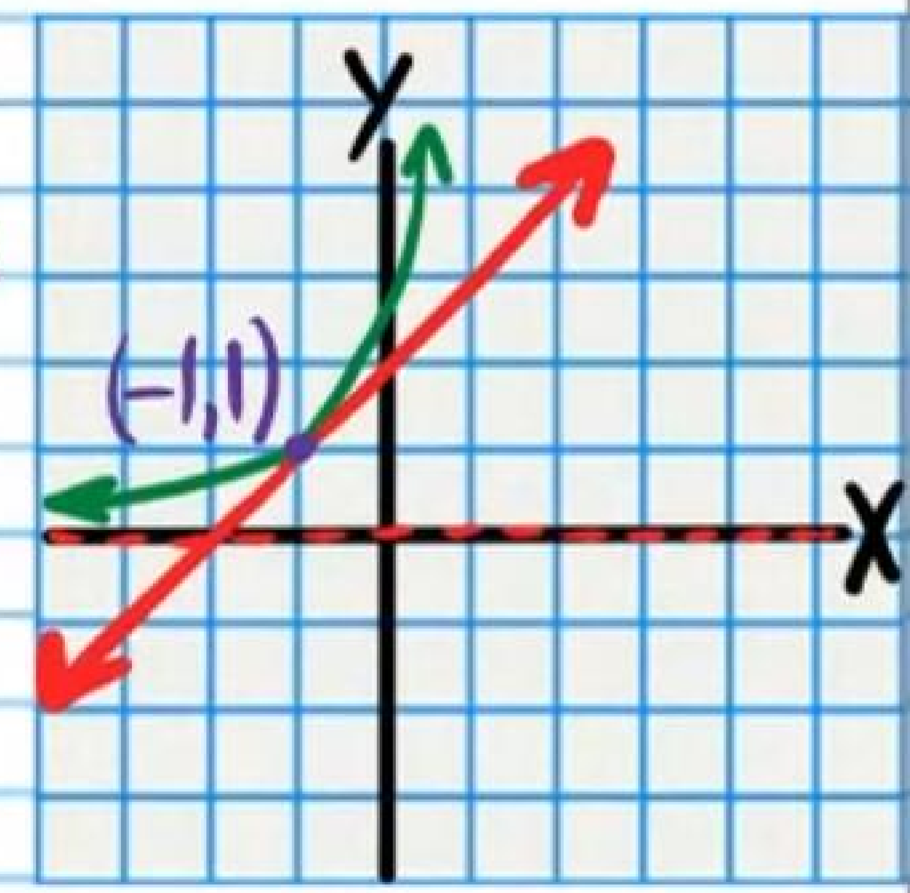
(23) $\ln y=x+1 \quad (-1,1)$

Finding the Slope of the Tangent Line

$$\frac{y'}{y}=1+0$$

$$y'=y$$

$$y'=1$$



Writing the Equation of the Tangent Line

$$y-y_p=m(x-x_p)$$

$$y-1=1(x-(-1))$$

$$\boxed{y-1=x+1}$$

Rule for Composite Natural Logarithm Functions

xiii) $\frac{d}{dx}\{\ln[g(x)]\} = \frac{g'(x)}{g(x)}$

- (24) $2x+1=\ln(4y^2-1)$, $\left(\frac{\ln 3-1}{2}, -1\right)$

Finding the Slope of the Tangent Line

$$2+0=\frac{8yy'}{4y^2-1}$$

$$2(4y^2-1)=8yy'$$

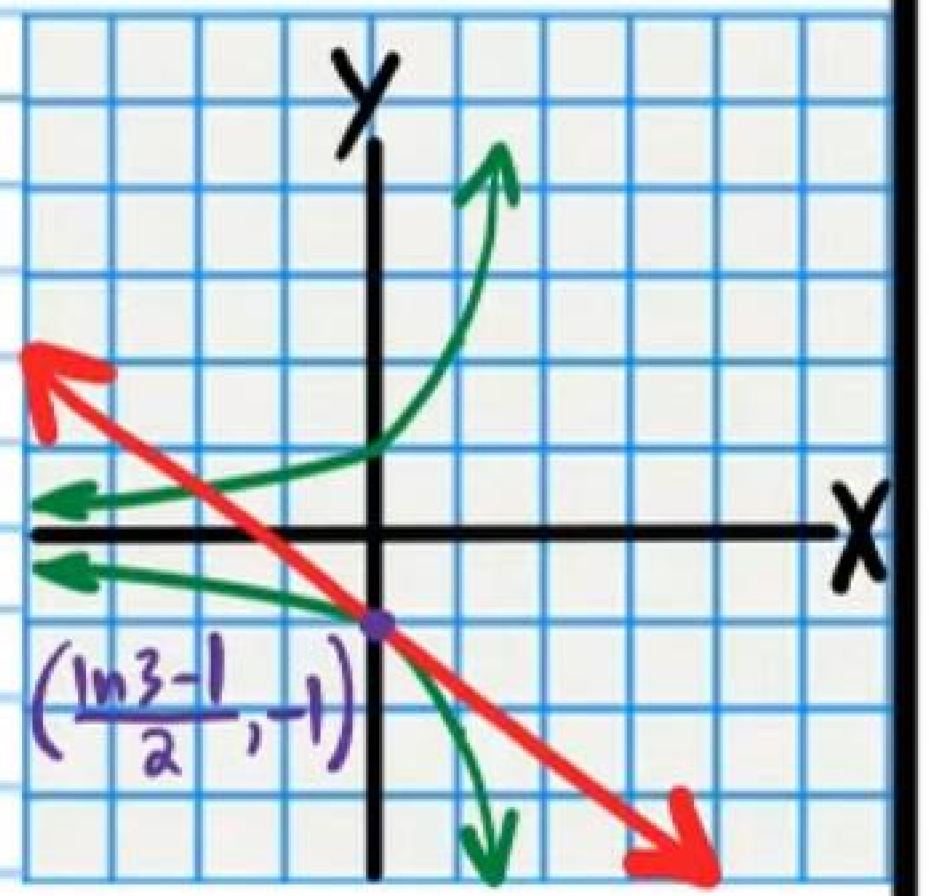
$$8y^2-2=8yy'$$

$$\frac{8y^2-2}{8y}=y'$$

$$y'=\frac{4y^2-1}{4y}$$

$$y'=\frac{4(-1)^2-1}{4(-1)}$$

$$y'=-\frac{3}{4}$$



Writing the Equation of the Tangent Line

$$y-y_p=m(x-x_p)$$

$$y-(-1)=-\frac{3}{4}\left(x-\frac{\ln 3-1}{2}\right)$$

$$\boxed{y+1=-\frac{3}{4}\left(x-\frac{\ln 3-1}{2}\right)}$$